How to Create a Bootable USB Drive for Windows

If you've ever needed to reinstall your operating system, fix a stubborn startup issue, or revive an older device, you've probably heard of creating an installation media or a bootable USB flash drive. It's an essential skill for anyone who manages or maintains computers.

While the process may seem technical, it's generally straightforward to follow, even for those with limited experience. In this guide, you'll learn what a bootable USB drive is, why it's useful, and how to create one for Windows 11. Whether you're prepping for a clean install or just keeping a backup handy, we'll walk you through everything: from choosing the right tools to running the commands.

Please note: Using a USB drive as bootable media in a way that is not supported by the manufacturer's warranty may result in that warranty being voided. Users should verify the warranty terms of their specific USB drive before attempting to use it as bootable media.

Disclaimer: The steps in this guide are provided for informational purposes only. Kingston does not support the creation or use of bootable USB drives. Kingston DataTraveler products are designed as storage devices, and using them as bootable media may lead to premature drive failure or other issues. Kingston does not provide technical support or warranty coverage for problems resulting from following these steps.

What is a bootable USB drive and what is it used for?

A bootable USB flash drive is a portable device that contains an operating system or recovery tool, which lets you start up (boot) a computer using the files stored on the USB, rather than the computer's built-in operating system. It's typically used for:

- **OS installation:** If you're setting up a new PC or reinstalling your OS, a bootable USB makes the process fast and easy.
- **System recovery:** When your operating system fails to load, you can boot from the USB and run repair tools.

- **Disk cloning:** A bootable USB can run disk cloning tools like **Acronis True Image**, allowing you to copy your existing HDD or SSD onto a new drive; perfect for **upgrading to a faster SSD** without reinstalling the OS or apps.

Whether you're an IT technician or a regular user, having a bootable USB can save time and data during emergencies.

How it works

When you create a bootable USB drive, you're writing a special ISO image (Windows) to write onto the USB drive in a way that makes it readable by your system's BIOS or UEFI firmware during startup. Once inserted into your computer and selected as the boot, your PC will treat the USB as if it were a physical installation disc or recovery tool.

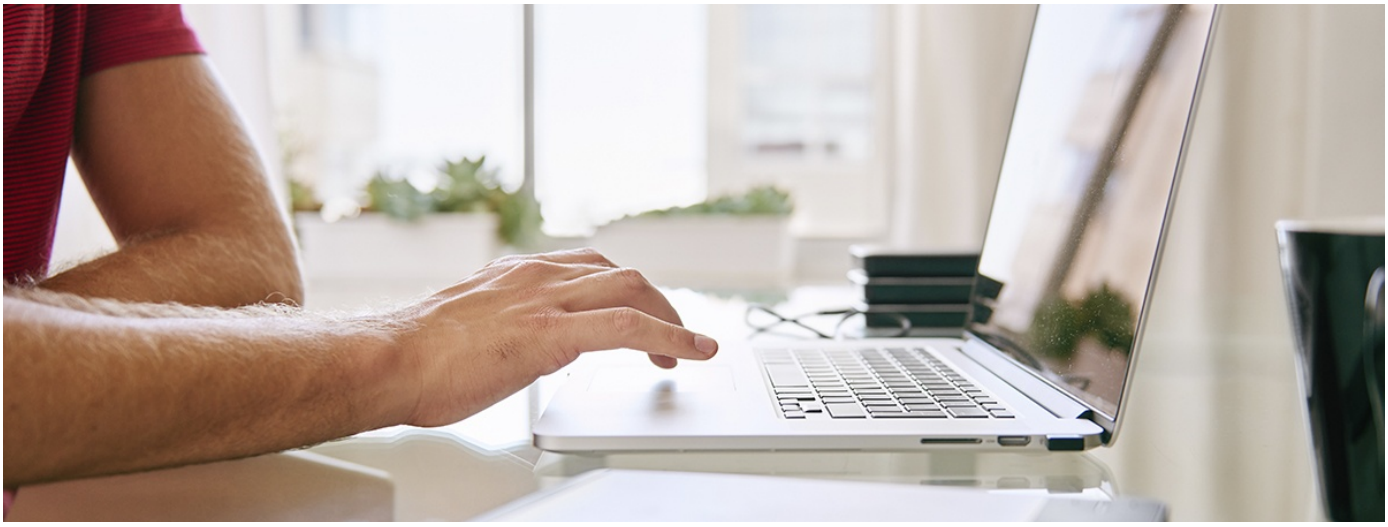
Creating a bootable USB drive

Creating a bootable USB drive is simpler than you might think. With the right tools, you can have one ready to go in just a few steps. Let us start with what you need, then we will show you how to set it up, and finally, make sure it works when you plug it in.

Step 1: Check what you'll need before you start

Before we show you how to make a bootable flash drive, you should double-check that you've got everything you will need, like:

- A USB flash drive with at least 8GB of space, but we recommend 16GB or more to ensure you have enough room for the installation files and any additional tools you might want to add.
- An OS Installation File, you can download the ISO files depending on what you're installing. For Windows 11, the official **Microsoft Media Creation Tool** is the easiest way to grab the ISO.
- Admin Access, as you'll need to be able to run system-level commands or install software. If you're not the admin on your device, ask someone who is to give you access.
- A stable internet connection to download the necessary installation files without interruptions.
- A computer running Windows to create the bootable USB drive from.



Step 2: Format the USB drive

Before you do anything, make sure there's nothing important on your USB flash drive, as the process will wipe everything clean. If y
got files you want to keep, now's the time to back them up. Once you're ready, go ahead and **format your USB drive**.

Step 3: Creating a Windows 11 bootable USB

Before we dive into the step-by-step instructions, it helps to know which tool is best for your needs. The options vary slightly depending on whether you're a beginner or advanced user, and whether you prefer Microsoft's official method or a third-party tool.

Here's a quick comparison between Microsoft's Media Creation Tool and a popular third-party app, Rufus, for creating bootable Windows USB drives:

Feature	Media Creation Tool	Rufus
Best For	Downloading and creating official Windows installation USBs	Creating bootable Windows USBs from existing ISO files
Target Users	Beginners, everyday Windows users	Advanced users who want customization options, IT professionals
Pros	• Official Microsoft tool • Automatically downloads the latest Windows ISO • Fast and easy to use	• Faster creation process • More customization (partition scheme, file system, etc.) • Works offline with pre-downloaded ISOs
Cons	• No advanced configuration options • Internet required • Slower USB creation	• Requires you to have or download the ISO separately • Interface may be less intuitive for beginners

Now let us go through both methods, starting with Media Creation Tool which is ideal for most users and then using Rufus which is ideal for more advanced users and those who want to customize their Windows.

Method 1: Creating a bootable USB using Microsoft's Media Creation Tool

If you're trying to install or reinstall Windows 11, the easiest method is by using Microsoft's official Media Creation Tool. Here is how to do it:

- Go to Microsoft's official **Windows 11 download page**.
- Scroll down to "**Create Windows 11 Installation Media**" and click "**Download Now**".
- Once the file is downloaded, open it and accept the license terms.
- Choose your preferred language and Windows edition.
- Select "**USB flash drive**" as the media type, then insert your pre-formatted USB.
- Select your USB from the list (click on "**Refresh drive list**" if it's not showing).
- Click "**Next**" and wait for the tool to download a Windows 11 copy to your USB.

Method 2: Creating a bootable USB using Rufus (more advanced)

1. **Download Windows ISO file**
   You can download the ISO directly from Microsoft's official **Windows 11 ISO download page** (scroll down to the "**Download Windows 11 Disk Image (ISO)**" section). Read the "Before you begin downloading an ISO" to ensure you have everything you need and your PC is compatible with Windows 11.
2. **Download Rufus**
   Head to the **Rufus website** and grab the latest portable version. It's a portable tool, so you won't need to install anything; all you need to do is run the .exe file.
3. **Plug in your USB drive**
   Insert your USB flash drive. Make sure there's nothing important on it, as the process will wipe the drive completely.
4. **Open Rufus**
   Double-click the executable to launch the app. It should automatically detect your USB device.
5. **Choose your ISO file**
   Under "Boot selection", click **SELECT** button, then locate and choose the Windows 11 ISO you want to use.
6. **Set your configuration**
   a. **Partition Scheme:** GPT (for UEFI) or MBR (for legacy BIOS/UEFI).
   b. **File System:** NTFS is recommended for Windows ISOs. Rufus will usually adjust these settings for you based on the ISO.
7. **Start the process**
   Click **Start**. You'll get a warning that the USB will be formatted, click confirm to continue. Rufus will begin writing the ISO to the drive.
8. **Finished**
   Once you see the "Ready" status, the USB is bootable.

Step 4: Test your bootable USB drive

Once the tool finishes, it's important to verify that your bootable USB works. This ensures everything was set up correctly and helps avoid surprises later when you actually need to use it. A quick check now can save time and frustration down the line.

Please note: The following steps involve entering your computer's BIOS/UEFI settings. Only proceed if you're familiar with or confident navigating these menus, as changing the wrong setting can cause startup issues. If you're not comfortable doing that, you can skip t[illegible] step, just remember you won't be able to confirm your USB is bootable until you actually use it.

- As your computer reboots, enter your BIOS/UEFI settings by pressing the key to access BIOS/UEFI (usually F2, Del, Esc, or F12 depending on your PC brand).
- In the boot menu or boot order settings, set the USB as first boot option by moving the USB drive to the top of the list.
- Then save your settings and restart. If the USB is bootable, you'll see the installer or boot environment start.

If it boots successfully, you know it works, you can now shut down the PC (Windows icon → Power icon → Shut down) and then remove the USB. **Do not remove or eject the USB while the system is running from it**, as it can crash the computer and may corrupt the data.

The next time you boot you may need to select your main internal drive in the BIOS/UEFI settings as described in step 2 through 4 above.

How to use a bootable USB drive

Using a bootable USB drive is simple once it's created. Plug the USB drive into the target computer and restart it. As the system begins to reboot, enter the BIOS or UEFI settings, just like you did during the testing phase by pressing the appropriate key for your system (commonly F2, Del, Esc, or F12). Once inside the BIOS/UEFI menu, navigate to the boot options and select your USB drive as the boot device.

The system should then start from the USB, launching the operating system or utility stored on it. Follow any on-screen instructions to begin installation, diagnostics, or whatever task you created the bootable drive for.

Bootable USB drives FAQs

Creating a bootable USB drive can raise a lot of questions, so we've gathered answers to the most common ones to help you troubleshoot and better understand the process.